EXP NO: 5	UNSUPERVISED LEARNING MODELS - CLUSTERING WITH K-MEANS

AIM:

To implement the K-Means clustering algorithm on the Iris dataset without using class labels, determine the optimal number of clusters, and evaluate and visualise the clusters.

ALGORITHM:

1. Import the required libraries (numpy, pandas, matplotlib, scikit-learn).
2. Load the Iris dataset and scale the features using StandardScaler (labels are not used for training).
3. Run K-Means for k = 2 to 10 and record the inertia and silhouette score for each k.
4. Plot the Elbow curve and the Silhouette curve to choose k.
5. Build the final K-Means model with k = 3 and obtain the cluster labels and centroids.
6. Evaluate the clusters using silhouette score and adjusted Rand index, and cross-tabulate clusters with species.
7. Reduce the data to 2 dimensions using PCA and plot the clusters with their centroids.

CODE:

```
# 1. Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score, adjusted_rand_score

# 2. Load dataset (labels are NOT used for training - unsupervised)
iris = load_iris()
X = iris.data
y_true = iris.target
X_scaled = StandardScaler().fit_transform(X)
print("Data shape:", X_scaled.shape)

# 3. Find the optimal number of clusters (Elbow + Silhouette)
inertia, sil = [], []
K = range(2, 11)
for k in K:
```

```
    km = KMeans(n_clusters=k, n_init=10, random_state=42).fit(X_scaled)
    inertia.append(km.inertia_)
    sil.append(silhouette_score(X_scaled, km.labels_))
    print(f"k={k}  Inertia={km.inertia_:.2f}  Silhouette={sil[-1]:.4f}")

plt.figure(figsize=(11, 4))
plt.subplot(1, 2, 1)
plt.plot(K, inertia, 'bo-'); plt.xlabel("Number of clusters (k)");
    plt.ylabel("Inertia")
plt.title("Elbow Method")
plt.subplot(1, 2, 2)
plt.plot(K, sil, 'gs-'); plt.xlabel("Number of clusters (k)");
    plt.ylabel("Silhouette Score")
plt.title("Silhouette Analysis")
plt.tight_layout()
plt.show()

# 4. Build final K-Means model with k = 3
kmeans = KMeans(n_clusters=3, n_init=10, random_state=42)
labels = kmeans.fit_predict(X_scaled)

# 5. Evaluate
print("\nCluster sizes:", np.bincount(labels))
print("Silhouette Score:", round(silhouette_score(X_scaled, labels), 4))
print("Adjusted Rand Index (vs true species):",
    round(adjusted_rand_score(y_true, labels), 4))
print("\nCluster Centers (scaled):\n",
      pd.DataFrame(kmeans.cluster_centers_,
          columns=iris.feature_names).round(3))
print("\nCluster vs Species cross-tab:\n", pd.crosstab(labels, y_true,
      rownames=['Cluster'], colnames=['Species']))

# 6. Visualise clusters using PCA (2-D)
pca = PCA(n_components=2)
X_2d = pca.fit_transform(X_scaled)
centers_2d = pca.transform(kmeans.cluster_centers_)
plt.figure(figsize=(7, 5))
plt.scatter(X_2d[:, 0], X_2d[:, 1], c=labels, cmap='viridis', s=40,
    edgecolor='k')
plt.scatter(centers_2d[:, 0], centers_2d[:, 1], c='red', marker='X', s=250,
    label='Centroids')
plt.xlabel("PCA 1"); plt.ylabel("PCA 2")
plt.title("K-Means Clustering (k=3) on Iris")
plt.legend()
plt.show()
```

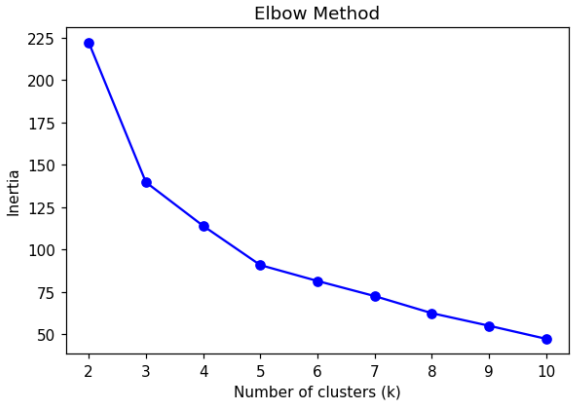
OUTPUT:

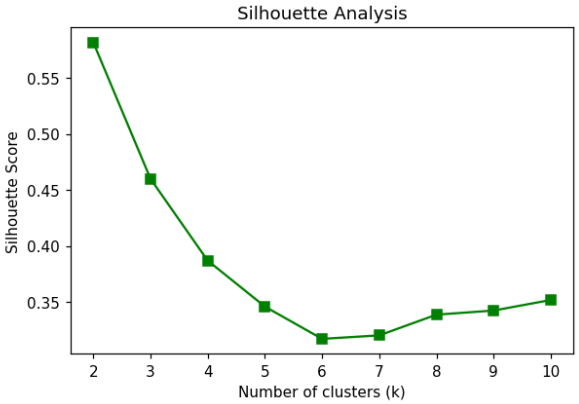
```
Data shape: (150, 4)
k=2  Inertia=222.36  Silhouette=0.5818
k=3  Inertia=139.82  Silhouette=0.4599
k=4  Inertia=114.09  Silhouette=0.3869
k=5  Inertia=90.93  Silhouette=0.3459
k=6  Inertia=81.54  Silhouette=0.3171
k=7  Inertia=72.63  Silhouette=0.3202
k=8  Inertia=62.54  Silhouette=0.3387
k=9  Inertia=55.12  Silhouette=0.3424
k=10  Inertia=47.39  Silhouette=0.3518

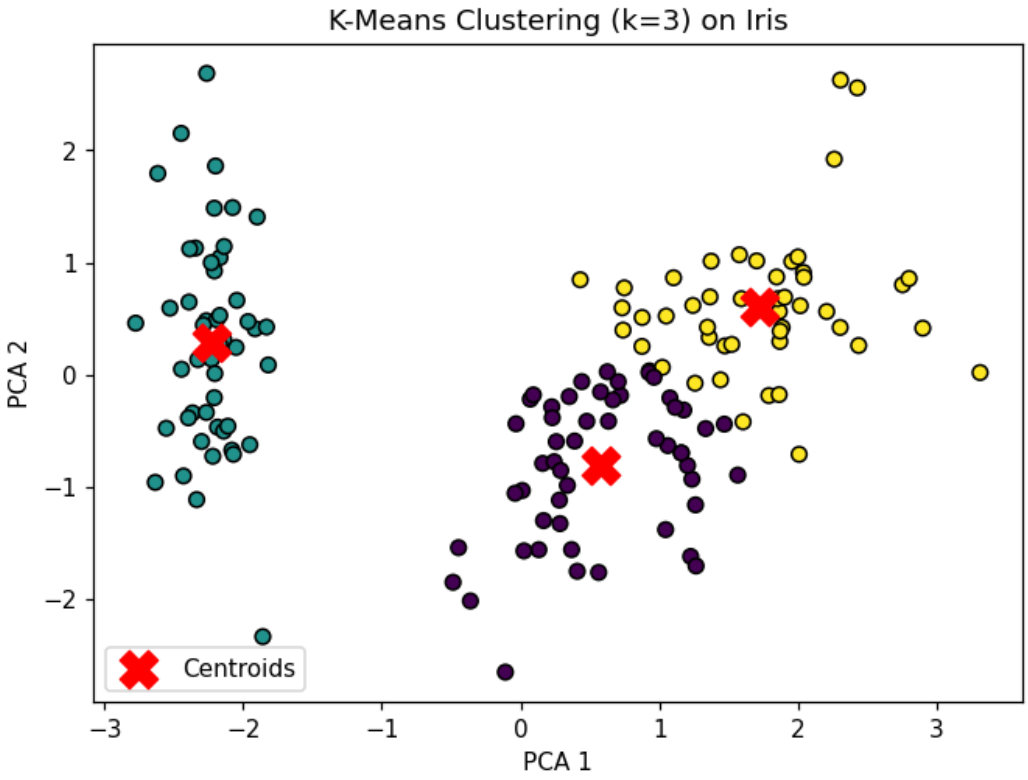
Cluster sizes: [53 50 47]
Silhouette Score: 0.4599
Adjusted Rand Index (vs true species): 0.6201

Cluster Centers (scaled):
    sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0              -0.050            -0.883              0.348             0.282
1              -1.015             0.853             -1.305            -1.255
2               1.136             0.088              0.996             1.018

Cluster vs Species cross-tab:
 Species   0   1   2
Cluster
0         0  39  14
1        50   0   0
2         0  11  36
```







RESULT:

K-Means clustering was successfully applied to the Iris dataset. With k = 3, the clusters had sizes of 53, 50 and 47, with a silhouette score of 0.4599 and an adjusted Rand index of 0.6201. The Setosa species was separated perfectly, while Versicolor and Virginica partially overlapped.